

Identifying Causal-Effect Inference Failure with Uncertainty-Aware Models

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NeurIPS 2020

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Story: a model makes a treatment recommendation

A hospital deploys a neural network to recommend cancer treatment.

- A new patient's profile — age, genetics, history — is unlike anyone in the training data.
- The model has seen no one like this under either treatment arm.
- The model outputs: **“Recommend treatment. Confidence: 94%.”**

The problem

Standard causal neural networks have no mechanism to say “I don't know.” They extrapolate silently and confidently into uncharted covariate space.

What is a causal neural network?

Standard ML: learn $f(x) \approx \mathbb{E}[Y \mid X = x]$.

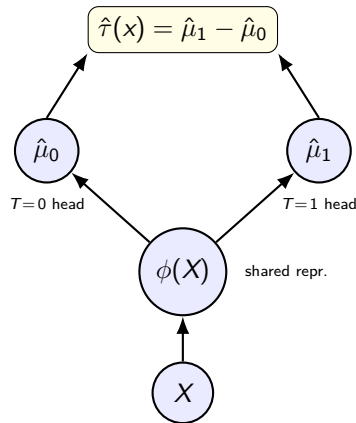
Causal ML: learn *two* response functions:

$$\mu_t(x) = \mathbb{E}[Y \mid X = x, T = t], \quad t \in \{0, 1\}$$

$$\hat{\tau}(x) = \mu_1(x) - \mu_0(x)$$

Key constraint: for each unit only *one* arm is observed ($T \cdot Y(1) + (1 - T) \cdot Y(0)$). The other is the missing counterfactual.

Architecture (TARNet): shared representation $\phi(X)$ followed by separate $T=0$ and $T=1$ prediction heads.



How are causal NNs trained?

Training is **fully supervised, end-to-end** — no unsupervised pre-training.

Factual loss (only the observed arm contributes per unit)

$$\mathcal{L}(\theta) = \sum_{i=1}^n \left[T_i \cdot \ell(Y_i, \hat{\mu}_1(\phi(X_i))) + (1 - T_i) \cdot \ell(Y_i, \hat{\mu}_0(\phi(X_i))) \right]$$

- Each unit activates *only one* head. ϕ is updated by backprop through the active head.
- The counterfactual head gets *no direct gradient signal* from that unit.

Consequence

ϕ is useful for factual prediction — but may not generalise to counterfactual prediction where no training counterparts exist in the opposite arm. **This is exactly the non-overlap problem.**

Course recap: CATE and the identification chain

Individual-level causal estimand:

$$\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x]$$

Under **unconfoundedness** $(Y(1), Y(0)) \perp\!\!\!\perp T \mid X$ and **consistency**:

$$\tau(x) = \underbrace{\mathbb{E}[Y \mid T = 1, X = x]}_{\mu_1(x)} - \underbrace{\mathbb{E}[Y \mid T = 0, X = x]}_{\mu_0(x)}$$

- $\mu_1(x)$ is estimated from *treated* training units near x .
- $\mu_0(x)$ is estimated from *untreated* training units near x .
- Both estimates require **positivity**: $0 < e(x) < 1$.

Positivity fails silently in high dimensions

Positivity assumption

$$0 < \mathbb{P}(T = 1 \mid X = x) < 1 \quad \text{for all } x \text{ in the support of } X.$$

- In low dimensions: checked with propensity score histograms.
- In high dimensions (images, EHRs, genomics): most patients are unique — positivity fails somewhere for almost everyone.
- Consequence: some $\mu_1(x)$ or $\mu_0(x)$ must be **extrapolated**, not estimated from nearby data.
- Standard causal neural networks extrapolate silently, with no signal to the user.

Classical fix: propensity score trimming

- 1 Estimate $\hat{e}(x) = \hat{\mathbb{P}}(T = 1 \mid X = x)$.
- 2 **Trim:** remove units where $\hat{e}(x) < \eta$ or $\hat{e}(x) > 1 - \eta$.
- 3 Estimate CATE only on the remaining data.

Intuition: units near $e(x) = 0$ have almost no counterparts in the $T = 1$ arm; units near $e(x) = 1$ have almost no counterparts in the $T = 0$ arm. Both make $\mu_1(x)$ or $\mu_0(x)$ unreliable.

The fundamental limit

The propensity $e(x)$ captures *marginal* treatment probability. It does not directly tell you *which arm* lacks data for a given x , nor by how much.

CEMNIST: a synthetic benchmark on real images

CEMNIST = Causal Effect MNIST — introduced by the authors as a new benchmark.

- **Covariates** X : real MNIST handwritten digit images (28×28 pixels, digits 0–9).
- **Treatment** T : binary, synthetically assigned — think of it as “received a new medication.”
- **Outcome** Y : binary, synthetically defined — think of it as “recovered.”
- **Why synthetic?** So the true CATE is *known by construction* and we can evaluate estimator quality.

This is a toy model

Treatment, outcome, and CATE values are *not real*.

The point is to create a controlled, high-dimensional setting where we can stress-test causal methods.

CEMNIST: data generating process

Treatment and outcome probabilities are defined *per digit class*:

Digit(s)	$p(x)$	$e(x) = \mathbb{P}(T=1 \mid x)$	$\mathbb{P}(Y=1 \mid x, T=0)$	$\mathbb{P}(Y=1 \mid x, T=1)$	CATE
9	0.50	$1/9 \approx 0.11$	1	0	-1
2	0.056	1.00	0	1	+1
other odd	0.056	0.50	1	0	-1
other even	0.056	0.50	0	1	+1

Digit 9 (50% of data): $e = 1/9$, so 8 in 9 nines are *untreated* and 1 in 9 are *treated*.
 $\Rightarrow \mu_0(9)$: estimated from many untreated nines.

$\Rightarrow \mu_1(9)$: estimated from some (fewer) treated nines.

Both arms have data. CATE is **estimable**.

Digit 2 (5.6% of data): *always* treated ($e = 1$).

No untreated twos exist in training.

$\Rightarrow \mu_0(2)$ must be **extrapolated**.

CATE is **not estimable**.

Why propensity trimming fails on CEMNIST

A trimmer sets thresholds η (lower) and $1 - \eta$ (upper):

- **Upper threshold** (e.g. $1 - \eta = 0.75$): removes units with $\hat{e}(x) > 0.75$. Digit 2 has $e = 1 \Rightarrow$ correctly trimmed. ✓
- **Lower threshold** (e.g. $\eta = 0.05$ – 0.10): removes units with $\hat{e}(x) < \eta$. Digit 9 has $e = 1/9 \approx 0.11$, and an estimated propensity model often pushes nines even closer to 0 — **nines get trimmed too**. ✗

Result: the trimmer removes both twos (correct) *and* nines (incorrect — nines have plenty of data on both arms).

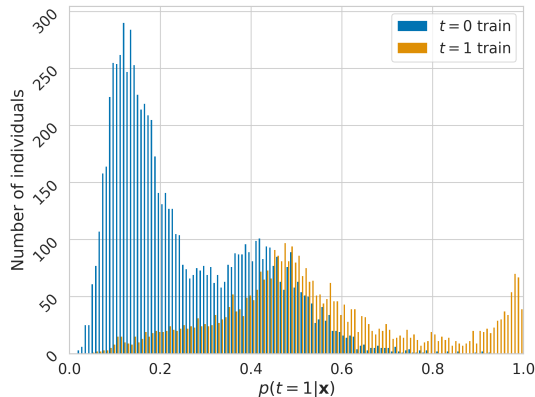


Fig. 2a: estimated propensity histogram. Large spike at left = untreated nines (many, low propensity). Trimming near $\eta \approx 0.10$ removes nines, not just twos.

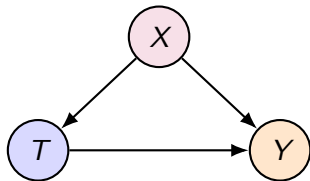
Core insight: non-overlap is covariate shift

Each arm model is trained on a *different* slice of the data: μ_0 on $p(X \mid T = 0)$, μ_1 on $p(X \mid T = 1)$. At test time we evaluate *both* at a new x .

Key observation

$e(x) \approx 0$: almost no treated unit like x in training $\Rightarrow x$ is OoD for μ_1 .

$e(x) \approx 1$: almost no untreated unit like x in training $\Rightarrow x$ is OoD for μ_0 .



Non-overlap = covariate shift

If overlap fails at x , at least one arm model is asked to predict outside its training support — the classical covariate shift problem.

This opens the door to OoD detection tools from ML.

Why does Bayes connect propensity to OoD?

We want to know: “Is x OoD for the $T = 1$ model?”

$\Leftrightarrow$ Is $p(x \mid T = 1)$ small?

Bayes' theorem

$$p(x \mid T = 1) = \frac{p(T = 1 \mid x) \cdot p(x)}{p(T = 1)} = \frac{e(x) \cdot p(x)}{p(T = 1)}$$

- $p(T = 1)$ and $p(x)$ are fixed population quantities, so $p(x \mid T = 1) \propto e(x)$.
- $e(x) \approx 0 \Rightarrow p(x \mid T = 1) \approx 0 \Rightarrow x$ OoD for μ_1 .
- $e(x) \approx 1 \Rightarrow p(x \mid T = 0) \approx 0 \Rightarrow x$ OoD for μ_0 .

Why this matters

$p(x \mid T)$ is not directly observable, but $e(x) = p(T = 1 \mid x)$ is estimable. Bayes makes OoD-ness **operationalizable** via the propensity score.

The problem and the paper's answer

Problem

Neural CATE estimators extrapolate silently when positivity fails. Propensity trimming uses a scalar $e(x)$ to decide estimability in a high-dimensional space — and gets it wrong (CEMNIST: removes estimable nines, keeps non-estimable twos).

Core insight

Non-overlap $\equiv$ covariate shift for one arm model:

$e(x) \approx 0 \Rightarrow x \text{ OoD for } \mu_1$ $e(x) \approx 1 \Rightarrow x \text{ OoD for } \mu_0$

Solution

Add epistemic uncertainty to the causal model (BNN / MC Dropout).

Uncertainty $U(x)$ is high exactly where OoD-ness occurs.

Reject on $U(x)$, not on $\hat{e}(x)$.

Epistemic vs. aleatoric uncertainty

Aleatoric uncertainty

Inherent noise in the outcome Y .

Example: even knowing the true $\mu_t(x)$ exactly, there is still randomness in Y .

Cannot be reduced with more training data.

Epistemic uncertainty

Uncertainty from *lack of training data* at x .

Example: x is OoD for μ_1 (digit 2 with $e = 1$) — many possible functions μ_1 fit the training data equally well.

Disappears as more relevant data arrives.

We want to detect

epistemic uncertainty

in $\hat{\tau}(x) = \hat{\mu}_1(x) - \hat{\mu}_0(x)$,

and use it to decide
when to withhold a recommendation.

Bayesian Neural Networks (BNNs): the tool

Replace a single fixed network $\mu_0(x; \omega_0)$ with a *distribution over networks*:

$$p(\omega_0 \mid \mathcal{D}) \quad (\text{posterior over weights given training data}).$$

Epistemic uncertainty = variance of predictions across BNN draws:

$$U_0(x) = \text{Var}_{\omega_0 \sim p(\omega_0 \mid \mathcal{D})} [\mu_0(x; \omega_0)].$$

- **In-distribution** x : all weight draws ω_0 give similar predictions $\Rightarrow$ low variance.
- **OoD** x : different ω_0 give wildly different predictions $\Rightarrow$ high variance.

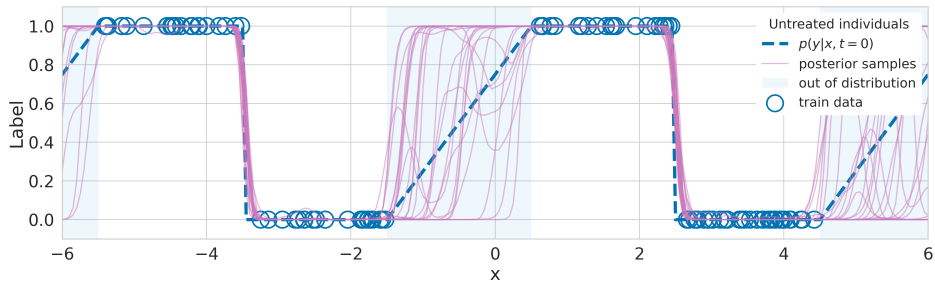
Practical approximation: MC Dropout (Gal & Ghahramani, 2016)

Add dropout to each outcome head; keep dropout *on* at test time.

Each stochastic forward pass $\approx$ one sample from $p(\omega \mid \mathcal{D})$.

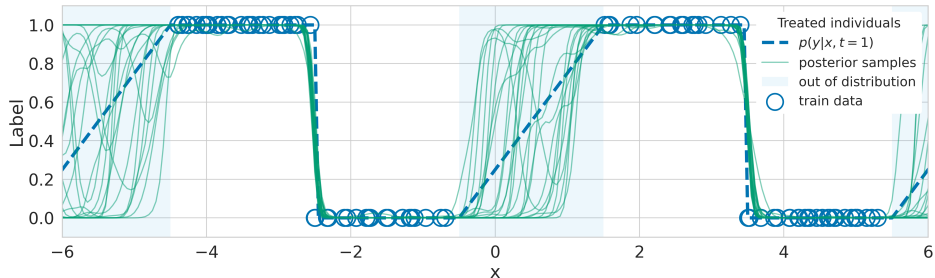
Run S passes; estimate $U(x)$ as the sample variance. Cheap.

Figure 1 (paper): BNN functions — $T = 0$ arm



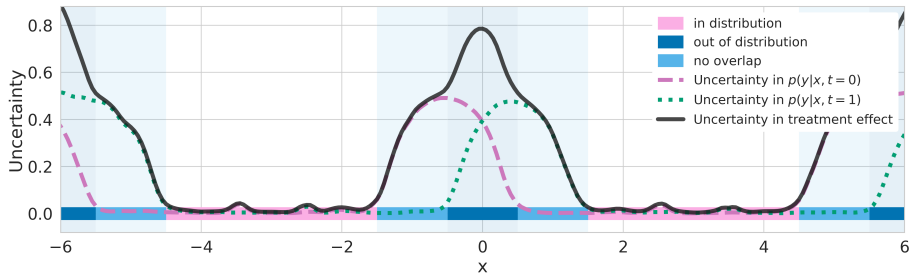
$T = 0$ arm. Each curve = one BNN weight draw ω_0 . **Left region** (shaded): $p(x | T = 0) > 0$ (many untreated units seen in training) $\Rightarrow$ all curves agree. **Right of shading**: $p(x | T = 0) \approx 0$ (OoD) $\Rightarrow$ curves diverge. High spread = high epistemic uncertainty $U_0(x)$.

Figure 1 (paper): BNN functions — $T = 1$ arm



$T = 1$ arm. **Right region** (shaded): $p(x | T = 1) > 0 \Rightarrow$ curves agree. **Left** of shading: low-propensity region ($e(x) \approx 0$, i.e. digit-2-like units). The μ_1 model has seen very few such units $\Rightarrow$ curves diverge, high $U_1(x)$.

Figure 1 (paper): CATE uncertainty



Bottom panel. Epistemic uncertainty in $\hat{\tau}(x)$ (solid black) is large *wherever at least one arm model is OoD* — non-overlap (light blue) or full covariate shift (dark blue). This is the signal the paper uses for rejection.

CATE uncertainty: step-by-step derivation

For a single BNN draw $\omega = (\omega_0, \omega_1)$, define $\hat{\tau}_\omega(x) = \mu_{\omega_1}(x) - \mu_{\omega_0}(x)$.

Apply the **law of total variance** conditioning on ω :

$$\begin{aligned}\text{Var}_{Y,\omega}[Y(1)-Y(0) \mid x] &= \underbrace{\mathbb{E}_\omega[\text{Var}_Y[Y(1)-Y(0) \mid \mu_{\omega_1}(x), \mu_{\omega_0}(x)]]}_{\text{aleatoric: noise in } Y \text{ even if we knew } \omega} \\ &+ \underbrace{\text{Var}_\omega[\mathbb{E}_Y[Y(1)-Y(0) \mid \mu_{\omega_1}(x), \mu_{\omega_0}(x)]]}_{\text{epistemic}}\end{aligned}$$

The epistemic term simplifies because $\mathbb{E}_Y[Y(t) \mid \mu_{\omega_t}(x)] = \mu_{\omega_t}(x)$:

$$\text{epistemic} = \text{Var}_\omega[\mu_{\omega_1}(x) - \mu_{\omega_0}(x)] = \text{Var}_\omega[\hat{\tau}_\omega(x)] =: U(x).$$

When is $U(x)$ large?

When different ω draws give very different $\hat{\tau}_\omega(x)$ — exactly when μ_{ω_1} or μ_{ω_0} is OoD at x .

Neural CATE estimators: a brief map

Model	Key idea	Bayesian extension
T-Learner	Fit μ_0, μ_1 separately	B-TLearner
TARNet	Shared representation, separate outcome heads	BTARNet
CFRNet	TARNet + MMD regularisation for overlap	BCFR-MMD
Dragonnet	TARNet + joint propensity prediction	BDragonnet
CEVAE	Latent confounders via VAE	BCEVAE + neg. sampling

Paper's recipe: add MC Dropout to the outcome heads of any model $\Rightarrow$ get $U(x)$ for free, with minimal code changes.

Rejection policy: thresholding on uncertainty

Setting: given budget r_{rej} , withhold the r_{rej} fraction with the worst signal. **Metric:** error rate and $\sqrt{\text{PEHE}}$ on non-rejected units.

	Uncertainty rejection	Propensity trimming
Rejection rule	$U(x) > \kappa$	$\hat{e}(x) < \eta$ or $> 1-\eta$
Signal type	model-based, arm-specific	marginal probability
Detects non-overlap	Yes, arm-by-arm	Yes
Detects covariate shift	Yes	No
CEMNIST: keeps twos?	No (high U , rejected)	No (high $\hat{e}$, rejected)
CEMNIST: keeps nines?	Yes (low U)	No (low $\hat{e}$, trimmed away)
Principled threshold	No closed-form rule	Caliendo & Kopeinig

Results: CEMNIST — uncertainty beats propensity

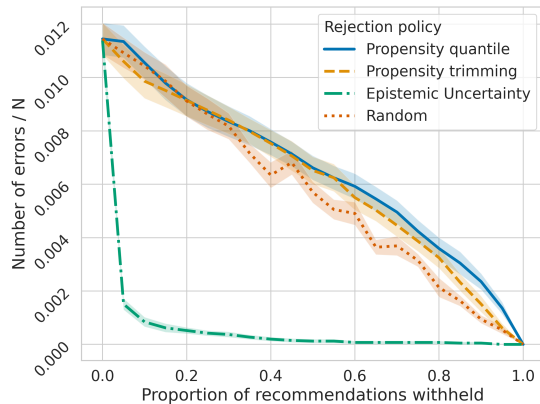


Fig. 2b: error rate vs. rejection rate. **Green dash-dot** (uncertainty) $\ll$ orange/blue (propensity) across the full range.

Results: IHDP and ACIC 2016

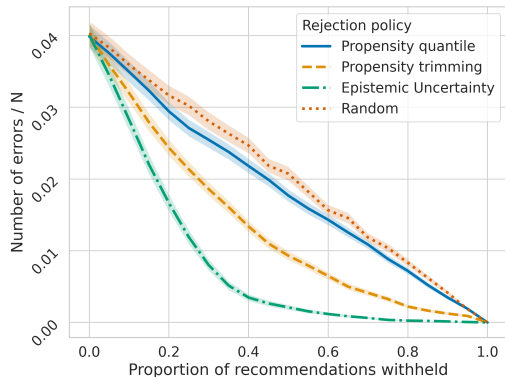


Fig. 3b: IHDP Covariate Shift. Uncertainty (green) outperforms propensity across all rejection rates.

Same result on:

- IHDP (no shift)
- ACIC 2016

All model classes improve.

Uncertainty is **more data-efficient**:
lower error at the same rejection rate.

What Jesson et al. (2020) show

- ① Non-overlap = covariate shift for one outcome model.
- ② Epistemic uncertainty in BNNs detects this — and detects input covariate shift as a bonus.
- ③ MC Dropout makes this cheap and practical.
- ④ Uncertainty-based rejection outperforms propensity trimming, especially when $e(x)$ and CATE estimability are decorrelated.

Now: how much should we trust this?

Assumption audit

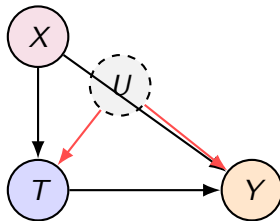
Assumption	Paper status	Notes
Unconfoundedness: $(Y(1), Y(0)) \perp\!\!\!\perp T \mid X$	Assumed, untested	Foundation of all CATE claims
SUTVA / consistency	Assumed, untested	Standard in literature
Positivity $0 < e(x) < 1$	Relaxed	The paper's contribution
Correct model class	Assumed	NN may misspecify μ_t
i.i.d. training data	Assumed	Violated under deployment shift

The paper relaxes positivity. It inherits all other classical assumptions intact.

The unconfoundedness blindspot

The method tells you **when the model is uncertain**. It does not tell you **whether the model is biased**.

If unconfoundedness fails — a hidden U confounds both T and Y :



- $\hat{\mu}_1(x)$ and $\hat{\mu}_0(x)$ both absorb the confounding signal.
- $U(x) = \text{Var}_\omega[\hat{\tau}(x)]$ can be small even when bias is large.

A Bayesian causal model can be *confidently wrong* under hidden confounding.

Two failure modes: only one is addressed

	Low bias	High bias (hidden confounding)
High $U(x)$	Reject $\Rightarrow$ safe.	Reject $\Rightarrow$ safe (by luck).
Low $U(x)$	Recommend $\Rightarrow$ safe.	Recommend $\Rightarrow$ confidently wrong. Blind spot.

The paper operates in the left column and top row.
The bottom-right cell is untouched.

What about sensitivity analysis?

Classical tools address hidden confounding directly:

- **Rosenbaum bounds:** how large must hidden U be to overturn the conclusion?
- **E-values:** minimum confounder strength to explain away the effect.
- **Random common cause refuters** (DoWhy): add a synthetic unobserved confounder; check if the estimate changes.

The paper has no sensitivity analysis.

It tells us *when* to trust a CATE estimate (overlap, in-distribution). It does not tell us *how robust* the estimate is to hidden confounding.

Uncertainty quantification and sensitivity analysis are **orthogonal**. A complete pipeline needs both.

The benchmarking problem

All experiments use **semi-synthetic** datasets:

- **IHDP**: real covariates from a clinical trial; synthetic outcomes.
- **ACIC 2016**: synthetic competition data; known ground truth.
- **CEMNIST**: MNIST digits; synthetic treatment and outcomes.

In semi-synthetic data, true CATE is known by construction $\Rightarrow \sqrt{\text{PEHE}}$ is computable.

In real-world observational data

$Y_i(0)$ for treated units is **never observed**.

$\sqrt{\text{PEHE}}$ is **uncomputable**.

The paper's main metric cannot be verified in deployment.

Comparison to classical methods

	Propensity trimming	Uncertainty rejection
Detects non-overlap	Yes, via $\hat{e}(x)$	Yes, more precisely
Handles covariate shift	No	Yes
Handles hidden confounding	No	No
Sensitivity analysis	Rosenbaum / E-values	None
Computationally cheap	Yes	No (BNN training)
Principled threshold	Caliendo & Kopeinig	No principled rule
Verifiable on real data	Yes	Yes, but $\sqrt{\text{PEHE}}$ uncomputable

What the paper genuinely contributes

- ① **Elegant reframing.** Non-overlap = covariate shift for one arm. Opens a toolkit (BNNs, OoD detection) not previously applied here.
- ② **Practical method.** MC Dropout adds uncertainty to any neural causal model with minimal engineering cost.
- ③ **New benchmark.** CEMNIST exposes a concrete failure mode of propensity trimming in high-dimensional, reproducible settings.
- ④ **Better rejection.** Outperforms propensity trimming on IHDP, ACIC, and CEMNIST across five model architectures.

What it does not contribute

A solution to hidden confounding, a sensitivity analysis, or a way to validate CATE estimates where ground truth is unavailable.

Discussion questions — Part A

- 1 **Uncertainty and confounding.** The paper assumes unconfoundedness throughout. If there are hidden confounders, can high $U(x)$ protect you? What would need to change to also catch confounding bias?
- 2 **Real-world validation.** The results are on semi-synthetic data. What experiment — using only observational data without known CATE — would convince you this holds in clinical practice?
- 3 **Redundancy with CFRNet.** CFRNet already penalises for overlap via MMD regularisation. Does MC Dropout on top give something genuinely new, or is it partially redundant?

- ④ **What has the model contributed?** You run the method and get high $U(x)$ for a patient. You withhold the recommendation and defer to a clinician. What has the causal model actually contributed? Is detecting uncertainty enough?
- ⑤ **Uncertainty about what, exactly?** The paper estimates $U(x) = \text{Var}_\omega[\hat{\tau}(x)]$. Is this the same as uncertainty about the true causal effect $Y(1) - Y(0)$? What components of total uncertainty are missing?

$$U(x) = \text{Var}_\omega [\hat{\mu}_{\omega_1}(x) - \hat{\mu}_{\omega_0}(x)]$$

- **What the paper gives you:** a principled signal for when to withhold a CATE estimate — detecting non-overlap and covariate shift simultaneously.
- **What it assumes and does not question:** unconfoundedness, correct model class, i.i.d. data. High $U(x)$ is not a guard against confounding bias.

The deeper lesson

Identification and uncertainty quantification are orthogonal.

You can know exactly how uncertain you are about the wrong estimand.
Epistemic uncertainty $\neq$ robustness to assumption violations.